

A LLM-based pedagogical framework for active, inquiry-based and adaptive learning in L2 writing

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ABSTRACT

Traditional L2 writing instruction often struggles to provide personalized, process-oriented feedback and engage student motivation. While generative AI like ChatGPT offers a potential solution, its application lacks a robust pedagogical foundation. This study proposes an innovative framework that integrates ChatGPT into L2 writing through a synthesis of active, inquiry-based, and adaptive learning principles. Within the framework, learners occupy the central position, undergoing the teaching implementation of LLM-based six-step writing instruction process to inquisitively learn, experiencing the teaching assessment of three-dimensional writing evaluation to actively acquire, and benefiting from the teaching reflection of plan for subsequent writing instruction to adaptively improve. Based on a quasi-experimental study involving 50 sophomores, the framework is proved to be effective, enhancing significantly learners' writing outcomes and motivation. These findings add to the limited body of research on the utilization of ChatGPT in education, providing valuable implications for research and pedagogical practices in L2 writing.

1. Introduction

With the emerging technology of large language models (LLMs) such as ChatGPT, language educators are prompted to rearrange their approaches to instructional design, assessment, and reflection. The pedagogical potential of LLMs in second language (L2) writing has generated significant scholarly interest (Li & Li, 2022; Pack & Maloney, 2024; Sarrion, 2023). Yet, while their usage is expanding, there remains a lack of robust conceptual frameworks to guide their integration into L2 pedagogy in a pedagogically meaningful way.

Existing studies have largely focused on teachers' pedagogical application of LLMs (Su et al., 2023; Özçelik and Ekşi, 2024), learners' perceptions of AI tools (Zou & Huang, 2023, 2024), and validation of LLM-based automated writing assessment (Bui & Barrot, 2024; Shin & Lee, 2023). However, a systematic framework for integrating LLMs into the full spectrum of L2 writing instruction to customize student learning experiences informed by active learning, inquiry-based learning (IBL), and adaptive learning, has yet to be developed.

These three pedagogical paradigms are foundational in modern education. Active learning emphasizes active learner participation in analysis, synthesis, and evaluation (Bonwell & Eison, 1991; Prince,

2004). Through active learning, as pointed out by Konopka et al. (2015), students can improve their understanding and retention of learnt knowledge, guaranteeing the shift of class focus from instructors towards students (DiYanni & Borst, 2020). IBL is an encompassing term of several pedagogical approaches which necessitate complete learner involvement in the whole learning process, with central focus on students' investigative work, raising questions and solving problems (Shah, 2020), for the purpose of developing students' metacognitive knowledge, high-order thinking, as well as self-regulated learning skills (Justice et al., 2007; Spronken-Smith & Walker, 2010). Adaptive learning uses technology to dynamically adjust course content level or type to tailor instruction to individual learner profiles (Capuano & Caballé, 2020). While each approach has shown value independently, their integration with AI tools like ChatGPT remains theoretically underdeveloped.

ChatGPT's potential to serve as a personalized learning assistant makes it well suited to support these pedagogical modes. Recent research illustrates how ChatGPT can enhance student engagement (Su & Lai, 2023), support writing skill development (Özçelik & Ekşi, 2024), and enable culturally responsive adaptive learning (Athanassopoulos et al., 2023). However, these efforts remain fragmented, lacking a

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cohesive framework to guide instructors in leveraging ChatGPT across all stages of instruction—design, delivery, assessment, and reflection.

To address this gap, this study proposes an integrative pedagogical framework for L2 writing instruction that aligns ChatGPT's affordances with the principles of active learning, IBL, and adaptive learning. The goal is twofold: (1) to delineate the epistemological mechanisms through which LLMs scaffold diverse pedagogical strategies, and (2) to provide practical implementation guidelines for educators in L2 writing instructional design, assessment, and reflection using AI tools.

This study addresses the following research questions: (1) How does ChatGPT interact with the principles of active learning, inquiry-based learning (IBL), and adaptive learning within the LLM-based pedagogical framework? (2) What instructional design principles and assessment strategies emerge as supporting productive interactions in this framework? (3) How effective is the proposed framework when implemented in real L2 writing contexts, as reflected in students' learning performance and motivation data?

To answer these questions, we adopt a three-part research design. First, we review the theoretical foundations of the three pedagogical approaches and examine existing AI integration practices. Second, we develop a conceptual framework that maps ChatGPT's capabilities onto core pedagogical processes. Finally, we validate the framework through an empirical case study in an authentic L2 writing classroom, examining its practical feasibility and impact on student learning outcomes.

2. Literature review

2.1. Active learning, inquiry-based learning and adaptive learning

According to Bonwell and Eison (1991), active learning, opposite to passive learning, referred to “the instructional activities involving students in doing things and thinking about what they are doing”. However, “certain specific obstacles are associated with the use of active learning, including a possible increase in preparation time, the potential difficulty of using active learning in large classes, and a lack of needed materials, equipment, or resources”. Prince (2004) then defined active learning as an encompassing term of discussions, case studies, and peer review and other teaching methods, with the purpose of promoting students' skill development and high-order thinking. DiYanni and Borst (2020) provided a more nuanced conceptualization of active learning, defining it as a pedagogical approach centered on sustained learner engagement throughout the learning process, rather than mere traditional lecture learning. Researches have been carried out on the effects of active learning strategies on students' learning, and positive effects are identified on both the learning outcomes (Khan & Madden, 2016; Nafea, 2020) and learning enthusiasm (Eustaqio et al., 2022). Through case analysis of applying speed learning in class, Khan and Madden (2016) revealed that active learning strategy can promote students' quick retention of learning materials at a relatively profound level, bringing enhanced learning outcomes in terms of learning efficiency. Meanwhile, Eustaqio et al. (2022) found that following the adoption of an active learning strategy within the educational settings, students' learning enthusiasm is greatly increased. It needs to point out that due to the limited groups of students being researched and over-reliance on student perceptions, the research findings are contextualized. Still, these findings prove that active learning strategies are effective due to their devotion to the student-centeredness, fostering adaptive environments to engage students in learning and move students from traditional knowledge acquisition to skill development and high-order thinking cultivation. In consideration of this, it is critical to perceive their potential for scaffolding IBL and adaptive learning. Although not explicitly pointed out in previous studies, the effectiveness of these strategies can be inferred to be partially attributed to learners' inquiring behaviors in the learning process and teachers' establishment of adaptive learning environments. In this regard, within our conceptual framework, we are targeted at addressing the dynamic interdependencies among active

learning, IBL, and adaptive learning. The framework posits that active inquiring learning process necessitate symbiotic integration with AI-facilitated adaptive scaffolding systems, which in this particular case is ChatGPT. This triadic pedagogical configuration engenders transformative potentialities for improving the learning experience and achievement of students in L2 writing domains.

For IBL, it is a pedagogical approach best involving learners in the knowledge creation process, with core attributes including student-centeredness (teacher as facilitator), self-directed learning, motivation of inquiry, acquisition based on seeking for new understanding, and an active approach to learning (Spronken-Smith, 2008). Levy et al. (2010) put it in a simple way, regarding IBL as a cluster of strongly student-centered approaches to learning and teaching that are driven by inquiry or research. Learners are required to take increased responsibility for their learning while seeking answers or solutions together with teachers and peers to the questions or problems posed in the learning activities. As suggested by Shah's research (2020), as students come across problem-solving tasks, they get to enhance many cognitive abilities, like logical presentation of arguments, critical reasoning, and creative thinking. The word ‘active’ thus implies learners' mobilization of high-order thinking to reconstruct knowledge system through engaging in inquiry activities in learning, like interpreting the data from experiment, analyzing a case study, or resolving a complicated problem from real world (Aditomo et al., 2011). As a consequence, IBL, which can be carried out through numerous approaches, is actually a fundamental part of active learning (Brew, 2003; Levy et al., 2010; Weaver, 1989).

As for adaptive learning, it is described as a learning model supplying appropriate learning content in accordance with the specific needs of an individual, which through utilizing the standards of learning technologies, can then be justified and evaluated (Yau & Joy, 2010). Yaghmaie and Bahreininejad (2011) further elaborated an individual's needs to include such factors as learners' characteristics, interest, and performance, thereby promoting adaptive learning as a process tailored to enhance both learning performance and learning satisfaction for distinctive learners. Thus, with the support of learning technologies, through automated and instructor interventions, not only is the course content type or level adjusted to enhance the individual learner's ability (Capuano & Caballé, 2020), but the whole learning itself serves to bring improved learner experience (Hwang et al., 2020; Ku & Sullivan, 2002).

Within the context of learner-centered education, prior research has demonstrated the positive effects of active learning, inquiry-based learning (IBL), and adaptive learning on learners' motivation, engagement, and overall outcomes throughout the learning process. Building on these findings, this paper proposes a large language model (LLM)-based pedagogical framework that integrates learner-centered strategies informed by active learning, IBL, and adaptive learning. A critical question is how educators and researchers can design effective interactions with LLMs grounded in IBL principles to enhance learners' motivation and engagement in active learning environments. Furthermore, it is important to consider how students' learning outcomes and preferences can be leveraged during the post-learning stage to guide reflective practices and facilitate adaptive learning through continued interaction with LLMs.

2.2. Applications of LLMs in L2 writing class

LLMs, with advanced computing and information processing techniques, are widely acknowledged to have the potential to transform the educational field (Firat, 2023; Huang & Li, 2023). Several researches have revealed LLMs' contributive impacts on L2 writing instruction by facilitating teaching design and implementation, automating assessment, as well as guiding material design (Hong, 2023; Rahman & Watanobe, 2023). Starting from learners' perspective, with the integration of LLMs, students' learning motivation, engagement and outcome can be enhanced (Baidoo-anu & Ansah, 2023; Hawanti &

Zubayduloevna, 2023; Imran & Almusharraf, 2023; Rahman & Watanobe, 2023). For learning motivation, Rahman and Watanobe (2023) pointed out that ChatGPT could be used to analyze learner performance, identify learner weakness, and monitor academic progress, which can then help customize learning design to satisfy the specific learning needs of individual students. Similarly, Hawanti and Zubayduloevna (2023), through comparing the levels of anxiety between two student writing groups, one group having access to ChatGPT and the other not, discovered that the group with access to ChatGPT show significantly lower anxiety levels. It is then summarized that ChatGPT integration in writing class could promote learner optimism. As such, students' learning motivation is naturally improved in the LLMs-aided learning environment. As for learning engagement, Baidoo-anu and Ansah (2023) indicated that through exploiting the ChatGPT's deep learning capacity, adaptive learning activities can be developed by instructors to suit the instructional approaches and content to students' levels, needs, and preferences for L2 learning. In other words, with students' learning preferences addressed by LLMs, students' learning engagement can be expected to increase in the learning process. Still, for learning outcome, Imran and Almusharraf (2023) examined the use of ChatGPT as a writing assistant in 30 articles, proving that LLMs can aid learners to generate drafts, brainstorm ideas, and organize research articles. Overall, LLMs can supplement class teaching, promoting students' active learning in aspects of learning motivation, engagement and outcome. Despite all these positive applications of LLMs in L2 writing in research and literature, it remains unclear how LLMs can be integrated in an all-round way, facilitating instructional planning and implementation, automating assessment, as well as material design, to bring out its full potential for L2 writing class. On the whole, here we first review previous studies in active learning, IBL and adaptive learning practice to

achieve an overall understanding of utilizing LLMs. On basis of this, we then develop a pedagogical framework positing LLMs in line with active learning, IBL, and adaptive learning. We hope that the proposed framework can help instructors to design teaching activities, so that a better interaction between learners with LLMs along with the emphasis on active learning and IBL can be ensured. Under this framework, it is also hoped that valuable insights can be gained into the role of LLMs in enhancing learning motivation, engagement and outcomes of students, hence guiding future theoretical development and empirical research.

3. A LLM-based pedagogical framework integrating ChatGPT for active, inquiry-based and adaptive learning in L2 writing

To fill in the gap mentioned above, this paper proposes a novel framework for L2 writing teaching. As shown in Fig. 1, the theoretical framework is comprised of an integral pedagogical paradigm encompassing three learning modes for teachers to carry out L2 writing teaching. The conceptual framework is structured for implementing LLMs, particularly ChatGPT, within formal accredited L2 writing instruction systems. The framework encompasses three core pedagogical components: Active Learning, IBL, and Adaptive Learning, adopting a learner-centered design that emphasizes the dynamic interplay between educators and AI technologies. This tripartite architecture positions human instructors and artificial intelligence systems as collaborative partners, working synergistically through continuous interaction with learners to facilitate the integration of the three pedagogical models within the educational ecosystem.

In the outermost layer, corresponding to principle behind the three learning modes, several strategic actions that a teacher can undertake when performing L2 writing teaching are provided. Arranged according

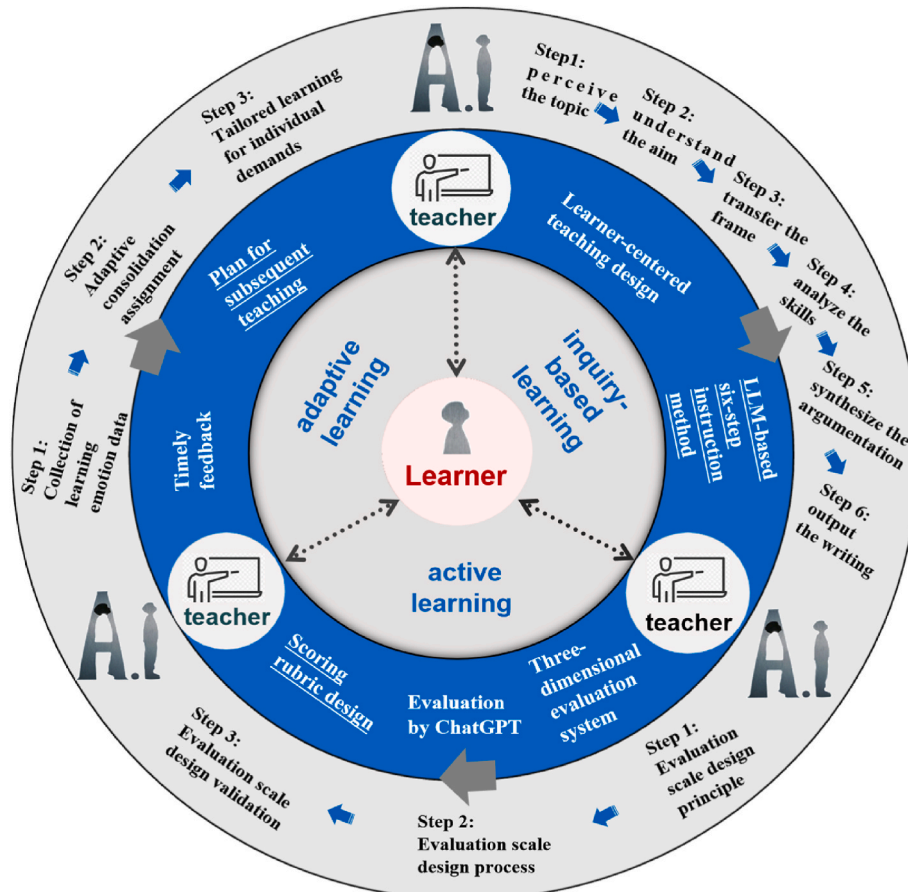


Fig. 1. The theoretical pedagogical framework integrating ChatGPT for active, inquiry-based and adaptive learning in L2 writing.

to the sequence of actions in normal teaching process, there are three major categories: 1) Teaching design and implementation for inquiry-based learning, including learner-centered teaching design, and LLM-based six-step instruction method; 2) Teaching assessment for active learning, covering three-dimensional evaluation system, evaluation by ChatGPT, and scoring rubric design; 3) Teaching reflection for adaptive learning, comprised of timely feedback and plan for second-round teaching. These actions may relate to one or multiple learning modes aligned with the learner-centeredness, acknowledging the inclusive and multifaceted nature of LLMs-based pedagogy. For instance, for “LLM-based six-step teaching method”, active learning, inquiry-based learning and adaptive learning are interwoven to guide learner's development of L2 writing ability. Similarly, for “Three-dimensional evaluation system”, active learning and inquiry-based learning are motivated for learner to explore deep into the complexity of teacher evaluation, peer evaluation and evaluation by ChatGPT.

Below, we describe the praxes in our framework that align with principle of the three learning modes as a whole.

3.1. Teaching design and implementation for inquiry-based learning

Teaching design and implementation, which are interrelated with each other, constitute the core of a practical pedagogical framework. Teaching design, or instructional design, according to Merrill et al. (1996), is a technology which incorporates known and verified learning strategies into instructional experiences which make the acquisition of knowledge and skill more efficient, effective, and appealing. On such theoretical basis and for the sake of differentiation, teaching design here refers to the overall ideology of incorporating strategies of active learning, inquiry-based learning, and adaptive learning for learner through introduction of ChatGPT, to address heterogeneous learner requirements in L2 writing education so as to improve their learning experience and proficiency. In contrast, teaching implementation denotes continuous, replicable steps for teachers to follow in utilizing ChatGPT to carry out concrete teaching activities in structured classroom environment.

3.1.1. Learner-centered teaching design

At the macro level, the whole teaching design is focused on learner, or to put it in a simpler way, learner-centered. Building upon Ouyang and Pengcheng's (2021) thorough investigation into the roles of AI technologies in education aligning with contemporary discourse on pedagogical theories, identified altogether three major paradigms: 1) AI-directed, learner-as-recipient paradigm; 2) AI-supported, learner-as-collaborator paradigm; and 3) AI-empowered, learner-as-leader paradigm. For this research, it needs to be pointed out that the idea of learner-centeredness corresponds to the third paradigm above, the AI-empowered, learner-as-leader paradigm. This theoretical positioning necessitates critical differentiation from preceding educational paradigms. In contrast to AI-directed, learner-as-recipient educational paradigm, where the learner acts as recipients to react to pre-designed sequences of knowledge, and implements pre-arranged series of learning activities set by AI for pre-supposed positive reinforcement to achieve pre-determined goals (Burton et al., 2004; Holmes et al., 2019), and the AI-supported, learner-as-collaborator educational paradigm, in which the learner acts as collaborators to interact with the AI systems in possession of their emerging, individualized information so as to realize better learning (Rose et al., 2019), the AI-empowered, learner-as-leader paradigm adopted in this framework emphasizes the persistent human-computer interactions during the prolonged learning process. In this way, regardless of the dynamic, complex nature of both the learner and AI systems, immediate data analysis as well as timely feedback can always be harnessed for the learner to improve the ongoing learning process. Learners get to take command of their own learning, controlling the risks of automatic AI decision and developing efficient learning. This learner-centeredness teaching design here truly demonstrates the

ultimate aim of AI application in education, which is augmenting the intelligence, capability, as well as potential of human beings (Tegmark, 2017).

3.1.2. LLM-based six-step instruction method

At the micro level, a LLM-based six-step teaching method for L2 writing is proposed, focusing on the writing process of learners. The writing process, defined as a functional dynamic system in which cognitive activities interact with together with the task environment and the writer's resources (van den Bergh et al., 2016), if used properly, can help improve the text quality in L2 writing (Al-Saadi & Galbraith, 2020; Zhu et al., 2023). In relating process to product, apart from engagement in planning processes and revision processes, the importance of the temporal distribution of the cognitive activities, or the moment when they take place, has also been discovered and proved (van den Bergh & Rijlaarsdam, 2001): Activities in the writing process, like reading and evaluation of the assignment, exerted a positive impact on the starting point of the writing process. During the writing process, a positive impact was identified for setting of goal, generation of idea and structuring arrangement. Vandermeulen et al. (2024) further deepened the research and verified the necessity of targeting effective process design by educators in order to offer concrete instruction to students, pointing out that an important choice for process instruction in L2 writing is aiding students in utilizing (AI) language tools strategically. In this way, they get to concentrate on planning of writing, translation of idea, and production of fluent text rather than on trivial problems at the word level, like spelling and grammar.

The writing processes discussed above are obviously closely related to Bloom's classic taxonomy on educational objectives, both attaching great significance to the cognitive domains of the learners in a sequential term. For Bloom's classic taxonomy on educational objectives, abbreviated as BCT, it identified 3 domains of learning (cognitive, affective and psychomotor domains) and was initially developed to assess students' learning outcomes through behavior observation (Bloom, 1956). For the purpose of the current research, we are exclusively devoted to the cognitive domain, in accordance with the focus of writing process or teaching procedure design. The cognitive domain is hierarchically classified into lower-order thinking skills (knowledge and comprehension), medium-order thinking skills (application and analysis), and higher-order thinking skills (synthesis and evaluation). As the BCT was originally proposed to describe and infer the cognitive achievement level, rather than select different sets of instructional activities or methods in the learning process (Athanasios et al., 2003), a revised Bloom's taxonomy (RBT) was proposed, retaining the core concept of BCT while transforming the “nouns” into “verbs”: remember, understand, apply, analyze, evaluate and create (Anderson & Krathwohl, 2001). To further expand the meaning of the different cognitive domains, unlike that of BCT, RBT consists of single words as well as their synonyms, allowing for more overlap and pedagogical activity use (Bümen, 2007).

In such regard, we chose RBT as our theoretical framework for a LLM-based six-step teaching method to guide students' learning process in L2 writing. As discussed earlier, in our incorporation of Bloom's revised taxonomy into the pedagogical framework, we have taken into consideration not only the keywords of the taxonomies, but also their synonyms. Fig. 2 illustrates the specific six-step instruction process of L2 writing supported by AI: 1) perceive the topic, 2) understand the aim, 3) transfer the frame, 4) analyze the skills, 5) synthesize the argumentation and 6) output the writing. Through these steps in progress, students are guided to gradually grasp the effective processes in fluent text production from theme understanding to theme extension, shifting truly their focus from the word level to the textual level in L2 writing.

For the first step “perceive the topic”, learners can prompt ChatGPT to “translate the topic”. Typical of LLMs, ChatGPT can not only give denotative meanings, translations and exemplar sentences as traditional printed dictionary or e-dictionary, but also simulate real-life scenarios to

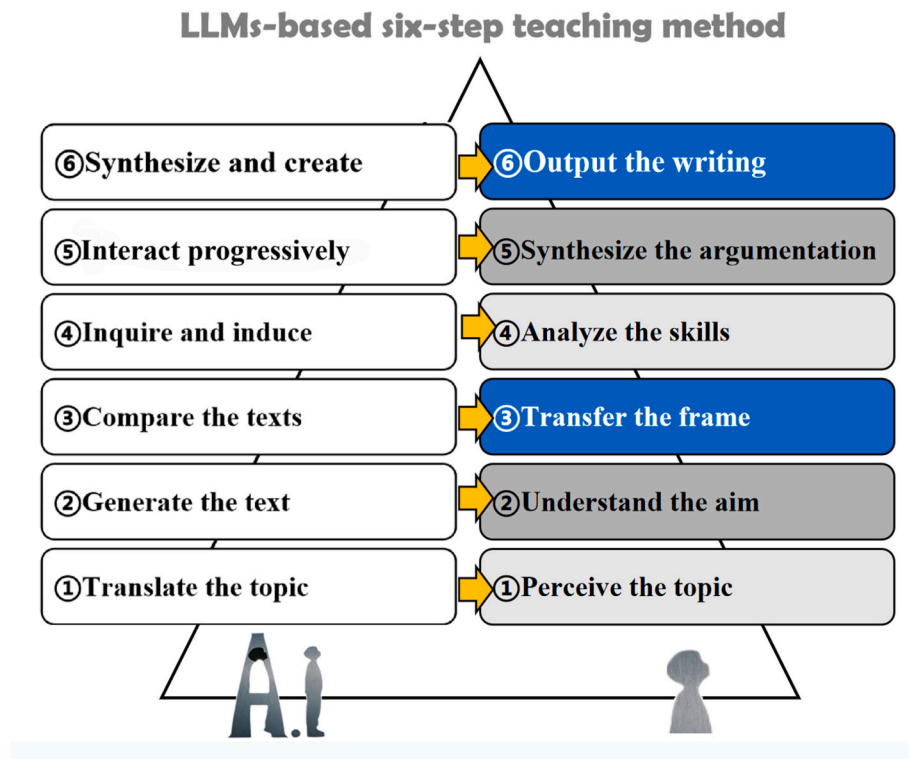


Fig. 2. A LLM-based six-step instruction process based on Blooms' Revised Taxonomy.

guide students perceive words in proper context (Kohnke et al., 2023). Besides, they can facilitate students' acquisition of targeted vocabulary through a theme (Sarrion, 2023) and adapt the dialogue sophistication to students' individual levels (Kohnke et al., 2023). Following this, students get to perceive the topic in a profound way. For the second step “understand the aim”, through exploiting the pre-trained dataset, ChatGPT can instantly “generate the text”, producing writing on the required topic with certain degree of logical coherence. If no further prompt is provided in a detailed manner, ChatGPT is liable to produce a loose argumentative essay. Thus, through comparing the loose writing version of ChatGPT and the rigid sample writing, students get to discover the aim of a particular writing genre. For the third step “transfer the frame”, learners can ask ChatGPT to simultaneously produce multiple writings, responding to the task requirements of relevance to topic, text genre, and word limit, so as to “compare the texts”. Through simple comparison, learners can easily deduce and transfer the general framework of the particular writing genre. For the fourth step “analyze the skills”, based on the multiple writings produced by ChatGPT above, learners can “inquire and induce” the writing skills behind different essay types. For the fifth step “synthesize the argumentation”, learners are guided to “interact progressively” with ChatGPT and gradually discover the specific writing methods utilized for argumentation. As for the sixth step “output the writing”, by installing proper real-life scenarios, learners are guided to synergize and create specific genre of essay adapted to real-life use. A diversity of writing tasks are designed for varied individual majors, interests and preferences. As shown above, these six steps together help facilitate the depth and pace of students' L2 writing learning. To facilitate understanding, the six-step instruction process regarding the specific genre of exemplification essay is illustrated in Appendix 1.

3.2. Teaching assessment for active learning

As an indispensable part of language teachers' work (Matsuda et al., 2011), writing assessment is rather complicated, covering different writing genres and types, devoting to both the learning process and

learning outcomes. Due to the over-emphasis on the mere summative learning achievement of assessment for the learning outcomes, researchers have gradually shifted to the research of assessment during the learning process, which is also called formative assessment or assessment for learning, since it can help promote L2 students' learning as well as L2 teachers' teaching through providing clearer and more consistent feedback for L2 writers to use language properly in L2 writing process (Hafner & Ho, 2020; Mak & Lee, 2014). Additionally, with the rapid development of AI, as pointed out by Li and Li (2022), it is necessary to further make use of the accessible technological tools to facilitate formative assessments so as to strengthen L2 writing learning. On such basis, exploiting the potential of AI as well as combining the advantages of traditional ways of assessment, this paper proposes a three-dimensional evaluation system, encompassing peer evaluation, evaluation by ChatGPT, and teacher feedback for the learning process.

3.2.1. Three-dimensional evaluation system

The overall evaluation system for the L2 writing is comprised of three parts: evaluation by ChatGPT, peer evaluation, and teacher feedback. For evaluation by ChatGPT, it serves to supplement and resolve the disagreement in peer evaluation result on the one hand, while relieve teachers' burden and lay a convincing foundation for teacher feedback on the other. Ever since Page (1966) proposed the concept of “computerized essay scoring” to relieve the burden of English teachers' grading responsibilities, researches on automatic essay scoring (AES) have been carried out, beginning with standard evaluation of academic writing (devoting exclusively to grammar, word choice and style) (Warschauer & Ware, 2006) to more inclusive evaluation of various writing genres (analyzing according to the particular parameters provided by the user) (Pack et al., 2024). Through evaluating the cross-sectional and longitudinal validity and reliability of LLMs, Pack et al. (2024) further validated the adoption of certain LLMs, especially ChatGPT, as a supplementary rater along with human raters or to resolve a disagreement between two human raters might be acceptable, as might their use in low-stakes or formative writing. In such regard, this paper proposes the integral evaluation system of evaluation by ChatGPT, peer

evaluation, and teacher feedback, together focusing on the promptness and accuracy in writing feedback, which in turn would bring improved learning experience of learners.

3.2.2. Evaluation by ChatGPT

Compared with traditional teacher feedback and peer evaluation, evaluation by ChatGPT exhibits significant advantage of promptness. Through entering a proper evaluation scale, ChatGPT can provide instant feedback on a particular writing output, including both numerical mark and human-like comment for revision and improvement. Additionally, the Generative AI systems are transformer-based deep neural network LLMs developed with substantial texts from the Internet (OpenAI, 2023). With the training text's domain-general nature, generative AI can actually interact with diversified genres of text, handling versatile scoring metrics and analyzing writing outputs according to whatever parameters being provided by a user. The combined merits of promptness and adaptability of evaluation by ChatGPT makes it an excellent choice for students in creating chances for more personalized, inclusive, and adaptable learning experiences, as well as for teachers in facilitating and transforming current educational practices (Holmes & Tuomi, 2022).

3.2.3. Scoring rubric design

With remarkable capacity in comprehending and generating meaningful content, LLMs' potential as an AES system has been recognized. Still, due to the preset database and generic working mechanism, LLMs may produce random feedback which reinforces preexisting biases, or even miss out on the nuances of a specific topic (Caines et al., 2023), decreasing in turn the validity and consistency of essay scoring by LLMs. For validity of essay scoring by LLMs, Parker et al. (2023) made a comparison between human raters and ChatGPT's scoring of nursing students' 42 essays, successfully proving the viability of scoring by ChatGPT. As for consistency of essay scoring by LLMs, Mizumoto and Eguchi (2023) leveraged GPT-3.5 model to carry out assessment of 12, 100 essays and proved a certain accuracy and reliability degree of ChatGPT in scoring essays, especially when linguistic features are specified in the evaluation rubric. Regarding this, it can be decided that a proper scoring rubric design is of critical significance for effective scoring by LLMs in L2 writing. Additionally, Hussein et al. (2019) indicated that LLMs overrides the limitation of AES systems in focusing solely on linguistic dimensions, and further addresses content-related dimensions, such as idea relevance to the prompt, development of idea, cohesion, and creativity.

In consideration of all these, combined with the learning objectives in L2 writing, the scoring rubric for ChatGPT is devoted to the assessment of students' writing texts so as to disclose their writing skill acquisition. These writing skills generally include the clear expression of thoughts, the logical organization of ideas, as well as the precise use of grammar, punctuation, and vocabulary, to ensure readable and effective writing output (Mohsen, 2022). Hence, the scoring rubric will be well designed to cover the four textual aspects of content, structure, language, and overall effect, as is shown in Appendix 2. Adapted from assessment standard for writing in College English Test in China, four scoring domains for argumentative essay in L2 writing are listed and explicated, with content centering on the writing skill of clear expression of thoughts, structure focusing on the skill of logical organization of ideas, language for the precise use of grammar, punctuation, and vocabulary, and overall effect for readable and effective writing output, which combined constitutes the rubric design principle. Following the first step of pinpointing rubric design principle, the scoring rubric still needs to go through two more procedures to be finally adopted in providing automatic scoring in class: the second step of rubric design process, which refers to the tailored design for the variety of writing genres, and the third step of rubric design validation, which refers to the testing and readjustment of scoring rubric to ensure scoring consistency and accuracy.

3.3. Teaching reflection for adaptive learning

For teaching reflection, it is an important move of teachers toward improved teaching design and implementation. Effective teaching reflection is based on the collection of timely feedback of students' learning and points to the tailored plan for second-round teaching, thus centering on students' adaptive learning.

3.3.1. Timely feedback

With proper scoring rubric provided, ChatGPT can give both scoring and descriptive feedback for the students' writing within a few seconds. As defined by Hattie and Timperley (2007), feedback is the information given to learners to think how to narrow or even eliminate the gap between a target performance and their actual performance, exerting positive effects on motivation, emotions, and learning itself. According to Shute (2008), feedback functions when it is timely and specific. For these, LLMs, with the numerous amounts of training texts to generate and manipulate human-like natural language, have already emerged as a potential solution in providing automated and concrete feedback to writing (Meyer et al., 2024). From the perspective of instructors, timely feedback is a reliable and fundamental source of students' learning data, including affective-motivational aspect and knowledge-skill aspect, which can be further exploited to readjust teaching plan and implementation in the second-round of teaching.

3.3.2. Tailored plan for subsequent teaching

With timely feedback of students' writing in the L2 writing learning process from LLMs by hand, instructors get to collect learners' learning emotion data, like learning interest of topics, learning preference of activities and learning motivation of multi-leveled writing tasks. Based on such data, teachers can release consolidation assignment adapted to students' learning interest, preference and motivation, improving students' subsequent learning. In the long term, tailored learning can be designed for the demands of individual students.

Overall, for effective teaching reflection and in turn tailored plan for second-round teaching, first of all, students' learning data should be well recorded; secondly, timely analysis of students' learning, including affective-motivational aspect as well as knowledge-skill aspect; thirdly, improved teaching design centering on students' adaptive learning should be proposed.

4. Evaluating the impact of pedagogical framework

A case study with quasi-experimental design was conducted to examine the effectiveness of the proposed LLM-based pedagogical framework, comparing (1) the L2 writing performance of an Experimental Group (EG) receiving the LLM-based instruction with that of a Control Group (CG) receiving conventional instruction, and (2) changes in students' learning motivation before and after the learning session. In educational research, quasi-experimental designs are widely used when random assignment is not feasible. As Abraham & MacDonald (2011) emphasize, such designs allow for the manipulation of an independent variable while accommodating real-world administrative constraints. In this study, intact classes were used, and a pretest-posttest control-group design was adopted.

Given that the pedagogical framework emphasizes active learning, IBL, and adaptive learning—processes highly dependent on learners' internal drive—not only learning achievement, but also motivation, the internal drive and desire of students to actively participate and persist in learning tasks (Emda, 2018), were assessed to provide a comprehensive evaluation of the framework's impact.

4.1. Methodology

4.1.1. Research design

As shown in Fig. 3, a complete 12-week learning session of College

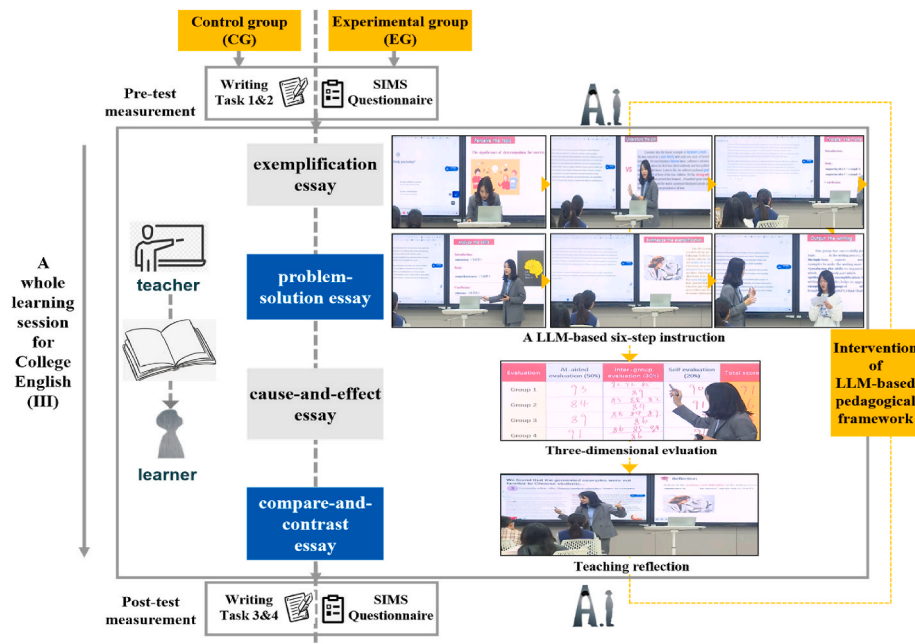


Fig. 3. A quasi-experimental research with pretest and posttest design for assessing impact of the LLM-based pedagogical framework.

English (III) was implemented for the two groups. Both EG and CG followed the same syllabus, course materials, instructor, and assessment schedule, but differed only in the pedagogical approach: EG received the LLM-based pedagogical framework introduced in Section 3, whereas CG received conventional instruction without LLM integration. To evaluate learning outcomes, two sets of instruments were used: (1) Argumentative writing assessments (pre- and post-tests) scored using carefully designed rubric (see Appendix 2); (2) The Situational Motivation Scale (SIMS) (Guay et al., 2000), assessing four motivational dimensions based on Self-Determination Theory (Deci & Ryan, 1985).

A two-way mixed-design ANOVA was applied, with Group (EG vs. CG) as a between-subjects factor and Time (pre-test vs. post-test) as a within-subjects factor. Assumptions of homogeneity were examined using Levene's test. Partial eta-squared (η^2) was reported as effect size, and Bonferroni-adjusted post hoc tests were performed where appropriate.

To complement quantitative findings, the instructor maintained structured observation logs and brief reflexive memos throughout the intervention. These qualitative accounts were later used to triangulate the quantitative data and strengthen the interpretation of the findings.

4.1.2. Participants

A total of 50 sophomores enrolled in College English (III) at a private university in Hubei Province participated in the study. Two intact classes of the same major with an identical mean age (around 19), a similar female-to-male ratio (approximately 1:3) and comparable English proficiency served as CG ($n = 24$) and EG ($n = 26$). Concerning the course, College English (III) is a mandatory course for students pursuing a bachelor's degree in China. It shifts from the focus of essential English reading and listening skills in College English (I) and College English (II) to applying English argumentative writing skills to solve real-world problems.

4.1.3. Procedures and instruments

With the pre-test set as the baseline, after a whole learning session with and without the intervention of the LLM-based pedagogical approach, the post-test results can directly show CG and EG students' L2 writing development and motivation change. To ensure reliable evaluation of students' argumentative writing performance: (1) The pre- and

post-tests each consisted of two parallel writing tasks directly selected from the College English Test Band 4 (CET-4) corpus by the National Education Examination Authorities or meticulously designed with equivalent difficulty level. Individual performance was then calculated as the mean of the rater's total scores across these two tasks, while the sub-scores for content, structure, language, and overall effect were derived in the same manner; (2) Tasks covered the four major argumentative genres: exemplification, problem-solution, cause-and-effect, and compare-and-contrast; (3) A certified CET-4 rater with extensive course experience evaluated all scripts using the rubric carefully designed mentioned in Section 3 (see Appendix 2) (content, structure, language, overall effect; 25 points each; total = 100; pass line = 60 points); (4) Parallel test design ensured equivalent difficulty and reduced memorization effects.

Construct validity was ensured by aligning task design with the 2020 College English Curriculum Requirements and by using authentic, real-world writing scenarios.

Concerning motivation assessment, Students' perceived motivational profiles were measured using the SIMS questionnaire, consisting of 16 items on a 5-point Likert scale measuring: Intrinsic Motivation (Q1-Q4), Identified Regulation (Q5-Q8), External Regulation (Q9-Q12), and Amotivation (Q13-Q16). This structure provides insight into the degree of self-determination driving students' engagement with L2 writing. According to Lepper et al. (1973), if the behavior is purely self-determined, there is a very high chance that it will be both intrinsically driven and that the behavior is done not for the reward or the prize, but rather for interest, enjoyment, and self-satisfaction, for the behavior itself. In such regard, as shown in Fig. 4, for intrinsic motivation, Q1-4 are respectively discussing the interest, enjoyment, and self-satisfaction. Q5-8 are devoted to the identified regulation of personal importance, consciousness, and valuing, Q9-12 are centered on external regulation of compliance and external rewards and punishments, and Q13-16 on amotivation of none-intentional, none-valuing, incompetence and lack of control. All the questions are on a 5-point Likert scale, with 1 indicating no correspondence at all, 2 indicating minor correspondence, 3 for medium correspondence, 4 for approximate correspondence, and 5 for exact correspondence.

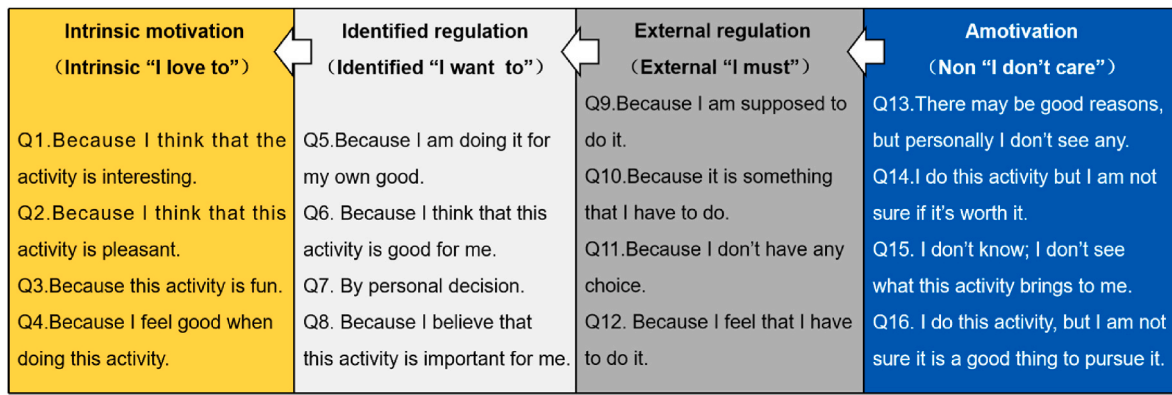


Fig. 4. An overview of the Situational Motivation Scale (SIMS) questionnaire.

4.1.4. Ethical considerations

To minimize bias and observer effects, students were not informed of their involvement in the study during the intervention. After the experiment, both groups were provided with equivalent instructional opportunities to ensure fairness. All procedures followed institutional guidelines for ethical educational research.

4.2. Results

4.2.1. Student's L2 writing performance

Descriptive statistics in Table 1 showed that both groups improved from pre-to post-test, with EG exhibiting substantially greater gains (Mean increase = 15.73) compared with CG (Mean increase = 6.79).

Levene's tests (see Table 2.) indicated homogeneity of variances for pre-test scores ($p = 0.879 > 0.05$), but not for post-test scores, indicating similar baseline performance across groups.

As shown in Table 3., a mixed-design ANOVA revealed that the main effect of Group was not significant with $F(1, 48) = 0.527$, $p = 0.471$, $\eta^2 = 0.011$, suggesting that the two groups did not differ in their overall mean performance. By contrast, the effect of Time was highly significant with $F(1, 48) = 228.632$, $p < 0.001$, $\eta^2 = 0.826$. According to the conventional benchmarks ($\eta^2 = 0.01 = \text{small}$, $\eta^2 = 0.06 = \text{moderate}$, $\eta^2 = 0.14 = \text{large}$), this represents a strong effect, suggesting that participants' scores improved from pre-test to post-test across groups. Importantly, a significant Time*Group interaction was observed, $F(1, 48) = 36.016$, $p < 0.001$, $\eta^2 = 0.429$, indicating that the improvement from pre-to post-test was significantly greater for EG compared to the CG. Taken together, these findings indicate that the observed improvement in the EG is not only statistically significant but also pedagogically meaningful.

To examine the specific effects of the innovative pedagogical framework on students' writing performance of content, structure, language, and overall effect, a two-way mixed-design ANOVA was further conducted for each subscale separately. As illustrated in Table 4, Time*Group interaction was significant ($p < 0.05$) for all the four subscales, revealing that EG achieved greater gains in content, structure, language, and overall effect performance than CG.

As illustrated in Fig. 5, EG participant A's exemplification essays in the pre- and post-tests are shown. In the pre-test, the participant exhibits obvious writing deficiency in content, structure, and language. Instead

Table 1

Pre-test and post-test L2 writing total scores for CG and EG.

Variable	Pre-test Mean	σ	Post-test Mean	σ	Mean difference	P-value
CG	68.46	17.17	75.25	17.26	6.79	0.17841
EG	67.12	15.98	82.85	10.70	15.73	0.00012

Note: σ = Standard Deviation.

Table 2

Levene's test of equality of error variances for CG and EG L2 writing total scores.

	Levene Statistic	df ₁	df ₂	P-value
Pre-test score	0.024	1	48	0.879
Post-test score	6.103	1	48	0.017

Table 3

Results of the two-way mixed-design ANOVA on L2 writing total scores.

Source	df ₁	df ₂	F	P-value	η^2
Group (between-subjects)	1	48	0.527	0.471	0.011
Time (within-subjects)	1	48	228.632	<0.001	0.826
Time * Group	1	48	36.016	<0.001	0.429

Note: η^2 = Partial eta-squared effect size.

of providing in-depth discussion of the main idea regarding social media and influencer marketing, participant A offers two homogeneous examples to show influencer marketing in a superficial way, leaving social media un-discussed. In structure there's a failure to offer a clear concluding sentence and language errors can be identified. After intervention, participant A shows an all-round writing improvement in content, structure, and language in the post-test.

In Fig. 6, EG participant B's problem-solution essays in the pre- and post-tests are shown. In the pre-test, the participant fails to make a correct judgment of the argumentative essay type and analyzes the pros and cons of online dating, rather than penetrating deep into the resolution of the problem. After intervention, participant B' writing shows obvious improvement in content and structure in the post-test, offering two solutions complementary to each other in the body part and evaluating the effect in the concluding part.

4.2.2. Student's motivation

As discussed earlier, to decide whether LLM-based pedagogical framework exerts a positive impact on student motivation, the SIMS questionnaire was sent to all the EG and CG participants before as well as after the learning session. Pre-test student motivation was measured before the learning session after taking the pre-test L2 writing tasks, while post-test student motivation was assessed after the learning session without taking the post-test L2 writing tasks.

To better visualize the comparisons made for the four categories of motivation assessed in the SIMS questionnaire, radar charts (see Fig. 7) are used, presenting the mean scores of Intrinsic Motivation (Q1-Q4), Identified Regulation (Q5-Q8), External Regulation (Q9-Q12), and Amotivation (Q13-Q16). The left panel displaying CG results shows a slight increase in intrinsic motivation and a small decrease in amotivation. The right panel for EG results demonstrates clear increases in

Table 4

Results of the two-way mixed-design ANOVA on subscales of L2 writing total scores.

Variable	CG		EG		ANOVA				
	Mean	σ	Mean	σ	Effect	F	df	P-value	η^2
Content									
Pre-test	16.33	4.584	15.85	4.324	Group	0.495	1	0.485	0.010
Post-test	17.75	4.504	19.81	2.562	Time	99.129	1	<0.001	0.674
					Time*Group	22.195	1	<0.001	0.316
Structure									
Pre-test	17.71	4.611	17.42	4.365	Group	0.648	1	0.425	0.013
Post-test	19.96	4.457	22.08	2.965	Time	161.88	1	<0.001	0.771
					Time*Group	19.625	1	<0.001	0.290
Language									
Pre-test	17.54	4.836	17.38	4.090	Group	0.269	1	0.607	0.006
Post-test	19.04	4.611	20.38	2.940	Time	65.218	1	<0.001	0.576
					Time*Group	7.246	1	0.01	0.131
Overall effect									
Pre-test	16.88	3.734	16.46	3.733	Group	0.675	1	0.415	0.014
Post-test	18.50	4.212	20.58	2.873	Time	175.81	1	<0.001	0.786
					Time*Group	33.09	1	<0.001	0.408

Pre-test (Task 1)

Directions: In the digital age, social media has emerged as a potent platform, bringing advertising success. Write a short essay on the impact of social media and influencer marketing with convincing examples.

Social Media and Influencer Marketing

In the digital age, social media has emerged as a potent platform for advertising, bringing advertising success. Consumers are influenced greatly by influencers.

My star Xiao Zhan is a famous Chinese singer and actor. In China, every young person knows him. He is the spokesman of many brands on TV. Because of his call, many people buy the goods. The sales of goods is improved because of Xiao Zhan. My girl star is Yang Zi. She introduced many goods in her TV series. When people watched her TV series, they would be attracted and bought the things they see in the TV. Today's advertising success is due to the famous stars. Because of our beloved idols, the businesses can have advertising success and make a lot of money.

Post-test (Task 3)

Directions: Is there a "success personality"--a special quality that leads to almost inevitably to achievement? Write a short essay on the significance of a particular personality with convincing examples.

The significance of challenges for success

The significance of challenges in achieving success is unmistakable, as illustrated by the remarkable journeys of both Thomas Edison and Zhang Zhongjing.

Edison, a towering figure in innovation, faced numerous obstacles throughout his career, most famously in his pursuit of inventing the electric light bulb. Despite encountering over a thousand failed attempts, Edison persisted, seeing each setback as an opportunity for growth and advancement rather than a roadblock. Similarly, Zhang Zhongjing, a revered figure in Chinese medicine, confronted significant challenges in his pursuit of medical knowledge during the Han dynasty. Despite living in a time of political unrest and limited resources, Zhang authored the influential work "Treatise on Cold Damage Disorders." His dedication to his craft, coupled with his ability to overcome adversity, enabled him to make enduring contributions to the field of medicine.

The examples of both Edison and Zhang underscore the universal acknowledgment of the importance of challenges in achieving success. Their experiences teach us that challenges are not barriers to be feared but rather opportunities for growth and innovation. By confronting challenges with resilience and determination, individuals can overcome obstacles and leave a lasting impact on their respective fields, just as Edison and Zhang did.

Fig. 5. Exemplification essay writings in pre-test and post-test for participant A in EG.

intrinsic motivation and identified regulation, along with noticeable decreases in external regulation and amotivation.

With Time set as within-subjects factor and Group as between-subjects factor, the two-way mixed-design ANOVA results for each subscale of intrinsic motivation (Q1-Q4), identified regulation (Q5-Q8), external regulation (Q9-Q12), and amotivation (Q13-Q16) were obtained. For intrinsic motivation (see Table 5), significant main effects of Time were observed across all four items ($p < 0.001$), indicating overall improvement from pre-to post-test. Moreover, the Group*Time interaction effects were all significant ($p < 0.001$), suggesting that EG experienced a greater increase in intrinsic motivation compared with CG. Specifically, for Q1 ($F(1, 48) = 60.591, p < 0.001, \eta^2 = 0.558$), Q2

($F(1, 48) = 53.675, p < 0.001, \eta^2 = 0.528$), Q3 ($F(1, 48) = 93.352, p < 0.001, \eta^2 = 0.660$), and Q4 ($F(1, 48) = 44.187, p < 0.001, \eta^2 = 0.479$), EG showed substantial post-test gains, whereas the CG demonstrated only minor changes. These results indicate that the LLM-based pedagogical framework was particularly effective in enhancing students' intrinsic motivation to learn.

Regarding identified regulation (see Table 6), a significant main effect of Time was also found for all items ($p < 0.001$), indicating improvement after the intervention. The Group*Time interaction was significant for Q5 ($p = 0.011$), Q6 ($p < 0.001$), Q7 ($p < 0.001$), and Q8 ($p < 0.001$), with medium-to-large effect sizes ($\eta^2 = 0.128-0.476$). These findings suggest that students in EG perceived greater value in the

Pre-test (Task 2)

Directions: With the fast development of online communication tools and online chatting tools, the **problem** of online dating has drawn widespread concern, requiring constructive solutions. Write a short essay on online dating.

The Pros and Cons of Online Dating

In the digital age, online dating has become increasingly popular. It offers several advantages. Firstly, it breaks through geographical limits, enabling people from different regions to connect. This greatly expands one's social circle. Secondly, the platforms provide a convenient way to communicate, allowing individuals to get to know each other at any time.

However, online dating also has its downsides. One major issue is the presence of deception. Some people may lie about their appearance, occupation, or background. Moreover, the virtual nature of online interactions can create a false sense of intimacy. People might develop strong feelings based on an idealized image, only to be disappointed when they meet in person.

In conclusion, while online dating offers opportunities for connection, users should be cautious and maintain a rational attitude to avoid potential risks.

Post-test (Task 4)

Directions: In recent years, accompanied by the widespread expansion of network, Internet fraud has become an increasingly serious **problem**. Write a short essay on Internet fraud.

How to solve the problem of Internet fraud?

Driven by a strong desire for high profits from various products, a number of online stores resort to shameful cheating. Some sell fake goods, some exaggerate the effect of the goods, and some deceive consumers with photos that do not match the goods. Internet fraud is emerging in an endless stream and has a detrimental impact on social integrity.

In order to prevent online fraud, I think at least two fundamental measures should be taken. To begin with, the **government** should establish even stricter laws and clearer regulations to ban Internet fraud. As for the **online shopping platforms**, like jd.com and Taobao, they should also shoulder their responsibilities, strengthening the supervision of the individual online stores on the one hand while organizing learning activities regarding social responsibility on the other, helping digital retailers to foster a strong sense of integrity.

Through these **measures**, online fraud will be effectively reduced and a more consumer-friendly online shopping environment will be built.

Fig. 6. Problem-solution essay writings in pre-test and post-test for participant B in EG.

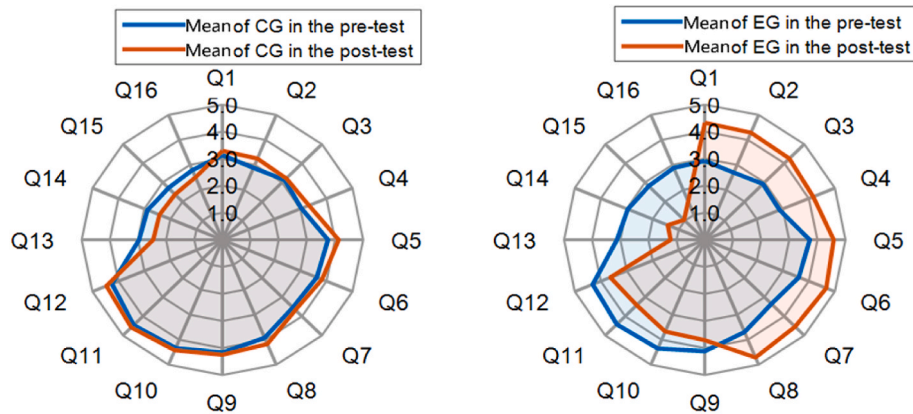


Fig. 7. The Impact of conventional approach and LLM-based pedagogical approach on individual motivation in L2 writing.

learning process and developed a stronger sense of personal identification with learning goals compared with CG.

Compared with intrinsic motivation and identified regulation, as shown in Table 7, external regulation showed weaker effects overall. The main effects of Time were not significant for most items (Q9, Q10, Q12; $p > 0.05$), with the exception of Q11 ($p = 0.004$). Importantly, significant Group*Time interactions were observed for Q10 ($F = 6.499$, $p = 0.014$, $\eta^2 = 0.119$), Q11 ($F = 15.493$, $p < 0.001$, $\eta^2 = 0.244$), and Q12 ($F = 7.062$, $p = 0.011$, $\eta^2 = 0.128$), while the Q9 interaction was not significant ($p = 0.124$). These interaction effects indicate differential change patterns between EG and CG. Inspection of the means revealed that EG's external regulation scores decreased from pre-to post-test (Q9: 4.12 $\rightarrow$ 3.81; Q10: 4.35 $\rightarrow$ 3.77; Q11: 4.42 $\rightarrow$ 3.50; Q12: 4.31 $\rightarrow$ 3.73),

whereas CG scores remained relatively stable or slightly increased. This pattern suggests that the intervention may have reduced students' reliance on external incentives or pressure, shifting motivational emphasis toward more self-determined forms.

For amotivation (see Table 8), significant Time effects were observed ($p < 0.001$, $\eta^2 = 0.688\text{--}0.768$), indicating an overall reduction in amotivation. Crucially, robust Group*Time interactions appeared for Q13 ($F = 51.121$, $p < 0.001$, $\eta^2 = 0.516$), Q14 ($F = 44.957$, $p < 0.001$, $\eta^2 = 0.484$), Q15 ($F = 65.572$, $p < 0.001$, $\eta^2 = 0.577$), and Q16 ($F = 42.657$, $p < 0.001$, $\eta^2 = 0.471$). EG demonstrated substantially larger decreases in amotivation than CG (Q13: 3.12 $\rightarrow$ 1.31; Q16: 2.88 $\rightarrow$ 1.58), implying that the intervention effectively mitigated feelings of disengagement or meaninglessness in learning.

Table 5

Results of the two-way mixed-design ANOVA on L2 writing intrinsic motivation.

Variable	CG		EG		ANOVA				
	Mean	σ	Mean	σ	Effect	F	df	P-value	η^2
Q1									
Pre-test	3.13	0.900	2.92	0.891	Group	2.481	1	<0.001	0.946
Post-test	3.29	0.955	4.23	0.710	Time	101.16	1	<0.001	0.678
					Time*Group	60.591	1	<0.001	0.558
Q2									
Pre-test	2.88	0.947	2.69	0.884	Group	2.633	1	0.111	0.052
Post-test	3.255	0.989	4.19	0.634	Time	149.10	1	<0.001	0.756
					Time*Group	53.675	1	<0.001	0.528
Q3									
Pre-test	3.13	0.741	2.92	0.935	Group	2.604	1	0.113	0.051
Post-test	3.21	0.833	4.15	0.834	Time	122.44	1	<0.001	0.718
					Time*Group	93.352	1	<0.001	0.660
Q4									
Pre-test	3.04	1.042	2.88	0.909	Group	1.333	1	0.254	0.027
Post-test	2.88	0.909	4.08	0.744	Time	119.96	1	<0.001	0.714
					Time*Group	44.187	1	<0.001	0.479

Table 6

Results of the two-way mixed-design ANOVA on L2 writing identified regulation.

Variable	CG		EG		ANOVA				
	Mean	σ	Mean	σ	Effect	F	df	P-value	η^2
Q5									
Pre-test	3.75	0.737	3.73	0.778	Group	0.565	1	0.456	0.012
Post-test	4.13	0.900	4.46	0.706	Time	68.197	1	<0.001	0.587
					Time*Group	7.059	1	0.011	0.128
Q6									
Pre-test	3.63	0.875	3.62	0.852	Group	2.521	1	0.119	0.050
Post-test	3.83	1.049	4.58	0.578	Time	75.078	1	<0.001	0.610
					Time*Group	31.122	1	<0.001	0.393
Q7									
Pre-test	3.54	0.779	3.35	0.846	Group	1.589	1	0.214	0.032
Post-test	3.67	0.917	4.42	0.758	Time	69.391	1	<0.001	0.591
					Time*Group	43.526	1	<0.001	0.476
Q8									
Pre-test	3.92	0.881	3.69	1.123	Group	0.329	1	0.569	0.007
Post-test	4.17	0.868	4.65	0.485	Time	45.181	1	<0.001	0.485
					Time*Group	15.584	1	<0.001	0.245

Table 7

Results of the two-way mixed-design ANOVA on L2 writing external regulation.

Variable	CG		EG		ANOVA				
	Mean	σ	Mean	σ	Effect	F	df	P-value	η^2
Q9									
Pre-test	4.17	0.637	4.12	0.653	Group	4.090	1	0.049	0.079
Post-test	4.25	0.608	3.81	0.567	Time	0.807	1	0.374	0.017
					Time*Group	2.451	1	0.124	0.049
Q10									
Pre-test	4.33	0.637	4.35	0.562	Group	6.162	1	0.017	0.114
Post-test	4.42	0.654	3.77	0.710	Time	3.632	1	0.063	0.070
					Time*Group	6.499	1	0.014	0.119
Q11									
Pre-test	4.46	0.588	4.42	0.703	Group	24.332	1	<0.001	0.336
Post-test	4.58	0.584	3.50	0.583	Time	8.984	1	0.004	0.158
					Time*Group	15.493	1	<0.001	0.244
Q12									
Pre-test	4.25	0.897	4.31	0.679	Group	4.341	1	0.043	0.083
Post-test	4.46	0.779	3.73	0.724	Time	1.556	1	0.218	0.031
					Time*Group	7.062	1	0.011	0.128

5. Discussion

The present study examined the effectiveness of a LLM-based pedagogical framework in enhancing students' L2 writing performance, relative to conventional instruction. Results from the mixed-design ANOVA revealed a significant main effect of Time, indicating that

both EG and CG improved from pre-test to post-test. Although no significant main effect of Group was found, the significant Group*Time interaction demonstrates that EG achieved substantially greater L2 writing gains than CG, highlighting the added value of the LLM-supported instructional approach. These performance-related findings were mirrored in the motivational data. Across the four motivational

Table 8

Results of the two-way mixed-design ANOVA on L2 writing amotivation.

Variable	CG		EG		ANOVA				
	Mean	σ	Mean	σ	Effect	F	df	P-value	η^2
Q13									
Pre-test	2.96	0.955	3.12	1.071	Group	3.836	1	0.056	0.074
Post-test	2.46	1.141	1.31	0.549	Time	159.20	1	<0.001	0.768
					Time*Group	51.121	1	<0.001	0.516
Q14									
Pre-test	2.87	0.680	2.96	0.720	Group	5.846	1	0.019	0.109
Post-test	2.42	0.830	1.46	0.508	Time	158.90	1	<0.001	0.768
					Time*Group	44.957	1	<0.001	0.484
Q15									
Pre-test	2.71	0.751	2.81	0.801	Group	9.873	1	0.003	0.171
Post-test	2.37	0.875	1.12	0.326	Time	145.69	1	<0.001	0.752
					Time*Group	65.572	1	<0.001	0.577
Q16									
Pre-test	2.79	0.721	2.88	0.766	Group	4.722	1	0.035	0.090
Post-test	2.50	0.885	1.58	0.504	Time	105.70	1	<0.001	0.688
					Time*Group	42.657	1	<0.001	0.471

dimensions—Intrinsic Motivation, Identified Regulation, External Regulation, and Amotivation—EG exhibited significant improvements in most subscales, particularly in intrinsic motivation (Q1-Q4) and identified regulation (Q5-Q8). In contrast, CG showed only minimal or non-significant changes across measures.

The divergence between EG and CG can primarily be attributed to the characteristics of the proposed pedagogical framework, which offered EG structured opportunities for inquiry, adaptive feedback, and iterative practice—conditions that were largely absent in CG's traditional classroom instruction. A secondary contextual factor relates to students' prior English writing training in Chinese high schools. Because CG participants had already acquired foundational writing skills, their progress may have plateaued, thereby limiting the magnitude of observable improvements within a short intervention period. Taken together, these results suggest that the LLM-based framework was the decisive factor driving the superior outcomes among EG participants.

The holistic improvements in EG students' L2 writing performance of content development, structural organization, linguistic accuracy, and overall effect can be explained by EG students' continuous exposure to well-formed English writing generated by ChatGPT, which likely strengthened their awareness of clarity in content expression and precision in language use—an advantage not available to CG students. Through iterative interaction with the LLM, EG participants gained opportunities to understand, analyze, and create written texts, enabling them to internalize the underlying logic of argumentative structure and produce more coherent essays. These findings are consistent with recent studies suggesting that interaction with ChatGPT enhances the structural, dialogic, and linguistic dimensions of argumentative writing (Su et al., 2023) and that LLMs demonstrate considerable potential in supporting logic development and L2 argumentative writing instruction (Zhang et al., 2025).

The motivational gains observed in EG can be further interpreted through the lens of Self-Determination Theory (Deci & Ryan, 1985). SDT posits that learners' intrinsic motivation and internalized regulation are strengthened when the learning environment supports their psychological needs for autonomy, competence, and relatedness. The current LLM-based framework—characterized by IBL, adaptive learning, and interactive engagement—appears to have fostered these needs more effectively than traditional instruction. The significant increase in intrinsic motivation indicates that EG students experienced the learning process as more enjoyable and meaningful, while the rise in identified regulation suggests a deeper appreciation of the personal value of L2 writing. Conversely, the limited or negative changes in external regulation and amotivation imply a shift away from controlled motivation toward more autonomous forms. This motivational shift is consistent with recent evidence showing that interaction with ChatGPT can

produce lasting improvements in learners' motivation to engage in English argumentative writing (Zare et al., 2025). Such increases in self-determined motivation likely supported the EG's superior performance, as intrinsic and identified regulation are well-established predictors of deeper cognitive engagement, sustained effort, and higher learning outcomes.

Overall, the convergence of performance, motivational, and qualitative writing improvements provides compelling evidence that the LLM-based pedagogical framework can serve as an effective and theoretically grounded approach to L2 writing instruction.

6. Conclusion

In this paper, a pedagogical framework is proposed, which encompasses the three distinct learning modes of active learning, IBL, and adaptive learning for individual learners through the integration of LLMs. With the learners centered upon, this framework puts great emphasis on the dynamic interplay among active learning, IBL, and adaptive learning. Within the triangular interactive mode among educators, LLMs and learners formed, we encompass the processes of teaching design and implementation, teaching assessment, and teaching reflection, enabling LLMs to fine-tune their responses and satisfy individual learner's needs. In the IBL, a learner-centric teaching design is incorporated with the pivotal component of a LLM-based six-step instruction method, while the active learning mode covers three-dimensional evaluation system, evaluation by ChatGPT and scoring rubric design for teaching assessment, and adaptive learning contains timely feedback and tailored plan for subsequent teaching for teaching reflection. These integral parts together form the basis underpinning active learning, IBL, and adaptive learning within the LLM-integrated educational context.

A quasi-experimental study further provided initial empirical support for the effectiveness of the proposed framework, with quantitative improvements in learners' L2 writing performance and motivation, plus qualitative insights gained from classroom observations. While the framework has shown promise in the context of L2 writing, its core principles may be adaptable to other disciplines and educational settings, showing potential for generalization.

Though we believe that the findings of the current research further validates the effectiveness of LLMs in scaffolding students' L2 writing, and adds to previous studies a novel pedagogical framework facilitating students' L2 writing from topic understanding of the writing task to the final argumentative writing output. Still, certain limitations warrant further consideration. Firstly, the experiment was carried out in one single institution with a relatively small sample size ($N = 50$), constraining the statistical power and generalizability of the findings

regarding students' L2 writing performance and motivation. Behind this, there was the consideration of balancing the time invested by school and students on the one hand and the labor devoted by researchers on the other. Ideally, this single-institution context along with the small sample size should be expanded to cover more students in multiple universities and colleges in China to further validate the effectiveness of the proposed pedagogical framework for L2 writing. Secondly, our findings were mainly related to students' L2 writing performance and learning motivation before and after the test over a relatively short span of time. The role of LLMs in assisting students over student learning was less clear, leading to potential bias. Future research therefore should conduct both writing process studies and longitudinal studies to examine the instant and sustained impact of LLM integration on learner performance and motivation. Finally, the research did not fully consider all the variables possibly exerting an effect on students' learning in the writing process, like the option of LLM platforms, learner demographics, learner proficiency level, and cultural background. Regarding implication for replication of the proposed pedagogical framework, it would be beneficial if future research could further evaluate how different LLM platforms (beyond ChatGPT) affect learning outcomes and user perceptions across diverse learners (beyond China).

To conclude, integrating LLMs into active learning, IBL, and adaptive learning shows enormous potential for improving both learner experience and performance. The pedagogical framework proposed offers an integrated approach to guide the design of LLMs in L2 writing instruction. Through in-depth exploration of the promise of LLMs technology and its capability to offer constant support, feedback, and guidance, instructors can foster more effectively engaging environments that promote active learning, IBL, and adaptive learning.

CRediT authorship contribution statement

Ruonan Wang: Writing – original draft, Methodology, Investigation.
Yan Yin: Writing – review & editing, Conceptualization. **Yongbo Cao:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.caeai.2025.100535>.

References

- Al-Saadi, Z. T., & Galbraith, D. (2020). Does revision process differ across language of writing (L1 vs. fl), language proficiency and gender? An empirical study using keystroke logging data. *Writing & Pedagogy*, 12(1), 73–109. <https://doi.org/10.1558/wap.38067>
- Anderson, L. W., & Krathwohl, D. R. (2001). *A taxonomy for learning, teaching and assessing: A revision of Bloom's taxonomy of educational objectives (completed.)*. Longman.
- Athanassiou, N., McNett, J., & Harvey, C. (2003). Critical thinking in the management classroom: Bloom's taxonomy as a learning tool. *Journal of Management Education*, 27(5), 533–555. <https://doi.org/10.1177/1052562903252515>
- Athanassopoulos, S., et al. (2023). The use of ChatGPT as a learning tool to improve foreign language writing in a multilingual and multicultural classroom. *Advances in Mobile Learning Educational Research*, 3(2), 818–824. <https://doi.org/10.25082/AMLER.2023.02.009>
- Bümen, N. T. (2007). Effects of the original versus revised bloom's taxonomy on lesson planning skills: A Turkish study among pre-service teachers. *The Review of Education*, 53, 439–455. <https://doi.org/10.1007/s11159-007-9052-1>
- Baidoo-anu, D., & Ansah, L. O. (2023). Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *iMeta*, 7(1), 52–62. <https://doi.org/10.4018/979-8-3693-9173-0.ch008>
- Bloom, B. (1956). *Taxonomy of educational objectives*. David McKay Company Inc.
- Bonwell, C. C., & Eison, J. A. (1991). *Active learning: Creating excitement in the classroom*. ASHE-ERIC Higher Education Report. George Washington University.
- Brew, A. (2003). Teaching and research: New relationships and their implications for inquiry-based teaching and learning in higher education. *Higher Education Research and Development*, 22(1), 3–18. <https://doi.org/10.1080/07294360.2012.642844>
- Bui, N. M., & Barrot, J. S. (2024). ChatGPT as an automated essay scoring tool in the writing classrooms: How it compares with human scoring. *Education and Information Technologies*, 30, 2041–2058. <https://doi.org/10.1007/s10639-024-12891-w>
- Burton, J. K., Moore, D. M. M., & Magliaro, S. G. (2004). Behaviorism and instructional technology. In D. H. Jonassen (Ed.), *Handbook of research on educational communications and technology* (2nd ed., pp. 3–36). Lawrence Erlbaum Associates Publishers.
- Caines, A., Benedetto, L., Taslimipour, S., Davis, C., Gao, Y., Andersen, O., et al. (2023). *On the application of large language models for language teaching and assessment technology*.
- Capuano, N., & Caballé, S. (2020). Adaptive learning technologies. *AI Magazine*, 41(2), 96–98. <https://doi.org/10.1609/aimag.v41i2.5317>
- Deci, E. L., & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. Plenum.
- DiYanni, R., & Borst, A. (2020). Active learning. In *The craft of college teaching* (pp. 42–60). Princeton University Press. <https://doi.org/10.23943/princeton/9780691183800.003.0004>
- Emda, A. (2018). Kedudukan motivasi belajar siswa dalam pembelajaran. *Lantanida Journal*, 5(2), 172–182. <https://doi.org/10.22373/lj.v5i2.2838>
- Eustasio, A., Gorráitz, M. da, & Lora, S. C. (2022). How the active learning method affects students' desire to learn. *International Journal of Curriculum Development, Teaching and Learning Innovation*, 1(1), 21–26. <https://doi.org/10.35335/curriculum.v1i1.58>
- Firat, M. (2023). What ChatGPT means for universities: Perceptions of scholars and students. *Journal of Applied Learning & Teaching*, 6(1), 1–22. <https://doi.org/10.37074/jalt.2023.6.1.22>
- Guay, F., Vallerand, R. J., & Blanchard, C. (2000). On the assessment of situational intrinsic and extrinsic motivation: The situational motivation scale (Sims). *Motivation and Emotion*, 24, 175–213. <https://doi.org/10.1023/A:1005614228250>
- Hafner, C. A., & Ho, W. Y. J. (2020). Assessing digital multimodal composing in second language writing: Towards a process-based model. *Journal of Second Language Writing*, 47, Article 100710. <https://doi.org/10.1016/j.jslw.2020.100710>
- Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, 77(1), 81–112. <https://doi.org/10.3102/003465430298487>
- Hawanti, S., & Zubayduloevna, K. M. (2023). AI chatbot-based learning: Alleviating students' anxiety in English writing classroom. *Bulletin of Social Informatics Theory and Application*, 7(2), 182–192. <https://doi.org/10.31763/businta.v7i2.659>
- Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education: Promises and implications for teaching and learning*. Center for Curriculum Redesign.
- Holmes, W., & Tuomi, I. (2022). State of the art and practice in ai in education. *European Journal of Education*, 57(4), 542–570. <https://doi.org/10.1111/ejed.12533>
- Hong, W. C. H. (2023). The impact of ChatGPT on foreign language teaching and learning: Opportunities in education and research. *Journal of Educational Technology and Innovation*, 5(1), 37–45. <https://doi.org/10.61414/jeti.v5i1.103>
- Huang, J., & Li, S. (2023). Opportunities and challenges in the application of ChatGPT in foreign language teaching. *International Journal of Education and Social Science Research*, 6(4), 75–89. <https://doi.org/10.1145/3700297.3700362>
- Hussein, M. A., Hassan, H., & Nassef, M. (2019). Automated language essay scoring systems: A literature review. *PeerJ Computer Science*, 5, Article e208. <https://doi.org/10.7717/peerj-cs.208>
- Hwang, G. J., Sung, H. Y., Chang, S. C., & Huang, X. C. (2020). A fuzzy expert system-based adaptive learning approach to improving students' learning performances by considering affective and cognitive factors. *Computers and Education: Artificial Intelligence*, 1. <https://doi.org/10.1016/j.caeai.2020.100003>. Article 100003.
- Imran, M., & Almusharraf, N. (2023). Analyzing the role of ChatGPT as a writing assistant at higher education level: A systematic review of the literature. *Contemporary Educational Technology*, 15(4), ep464. <https://doi.org/10.30935/cedtech/13605>
- Justice, C., Rice, J., Warry, W., Inglis, S., Miller, S., & Sammon, S. (2007). Inquiry in higher education: Reflections and directions on course design and teaching methods. *Innovation in Higher Education*, 31, 201–214. <https://doi.org/10.1007/s10755-006-9021-9>
- Khan, A., & Madden, J. (2016). Speed learning: Maximizing student learning and engagement in a limited amount of time. *International Journal of Modern Education and Computer Science*, 8, 22–30. <https://doi.org/10.5815/ijmecs.2016.07.03>
- Kohnke, L., Moorhouse, B. L., & Zou, D. (2023). ChatGPT for language teaching and learning. *RELC Journal*, 54(2), 537–550. https://doi.org/10.1007/978-3-031-51447-0_97-1
- Konopka, C. L., Adaime, M. B., & Mosele, P. H. (2015). Active teaching and learning methodologies: Some considerations. *Creative Education*, 14, 1536–1545. <https://doi.org/10.4236/ce.2015.614154>

- Ku, H. Y., & Sullivan, H. J. (2002). Student performance and attitudes using personalized mathematics instruction. *Educational Technology Research & Development*, 50(1), 21–34. <https://doi.org/10.1007/BF02504959>
- Lepper, M., Greene, D., & Nisbett, R. (1973). Undermining children's intrinsic interest with extrinsic rewards: A test of the "overjustification" hypothesis. *Journal of Personality and Social Psychology*, 28, 129–137. <https://doi.org/10.1037/h0035519>
- Levy, P., Little, S., McKinney, P., Nibbs, A., & Wood, J. (2010). *The Sheffield companion to inquiry-based learning*. The University of Sheffield.
- Li, J., & Li, M. (2022). Assessing L2 writing in the digital age: Opportunities and challenges. *Journal of Second Language Writing*, 57, Article 100679. <https://doi.org/10.1016/j.jslw.2022.100913>
- Mak, P., & Lee, I. (2014). Implementing assessment for learning in L2 writing: An activity theory perspective. *System*, 47, 73–87. <https://doi.org/10.1016/j.system.2014.09.018>
- Matsuda, P. K., Cox, M., Jordan, J., & Ortmeier-Hooper, C. (Eds.). (2011). *Second-language writing in the composition classroom: A critical sourcebook*. Bedford/St. Martin's.
- Merrill, M. D., Drake, L., Lacy, M. J., & Pratt, J. (1996). Reclaiming instructional design. *Educational Technology*, 36(5), 5–7.
- Meyer, J., Horbach, A., Jansen, T., Schiller, R., & Fleckenstein, J. (2024). Using LLMs to bring evidence-based feedback into the classroom: AI-generated feedback increases secondary students' text revision, motivation, and positive emotions. *Computers and Education: Artificial Intelligence*, 6, Article 100199. <https://doi.org/10.1016/j.caeai.2023.100199>
- Mizumoto, A., & Eguchi, M. (2023). Exploring the potential of using an AI language model for automated essay scoring. *Research Methods in Applied Linguistics*, 2(2), Article 100050. <https://doi.org/10.1016/j.rmal.2023.100050>
- Mohsen, M. A. (2022). Computer-mediated corrective feedback to improve L2 writing skills: A meta-analysis. *Journal of Educational Computing Research*, 60(5), 1253–1276. <https://doi.org/10.1177/07356331211064066>
- Nafea, I. (2020). Active learning multi-strategies. *International Journal of Innovative Technology and Exploring Engineering*, 10(1), 87–94. <https://doi.org/10.35940/ijitee.A8119.1110120>
- Özçelik, N. P., & Ekşi, G. Y. (2024). Cultivating writing skills: The role of ChatGPT as a learning assistant—a case study. *Smart Learning Environments*, 11(10), 1–18. <https://doi.org/10.1186/s40561-024-00296-8>
- OpenAI. (2023). Introducing GPTs. <https://openai.com/blog/introducing-gpts>. (Accessed 6 November 2023).
- Ouyang, F., & Pengcheng, J. (2021). Artificial intelligence in education: The three paradigms. *Computers and Education: Artificial Intelligence*, 2, Article 100020. <https://doi.org/10.1016/j.caeai.2021.100020>
- Pack, A., Barrett, A., & Escalante, J. (2024). Large language models and automated essay scoring of English language learner writing: Insights into validity and reliability. *Computers and Education: Artificial Intelligence*, 6, Article 100234. <https://doi.org/10.1016/j.caeai.2024.100234>
- Pack, A., & Maloney, J. (2024). Using artificial intelligence in TESOL: Some ethical and pedagogical considerations. *Tesol Quarterly*, 58, 1007–1018. <https://doi.org/10.1002/tesq.3320>
- Page, E. B. (1966). The imminence of grading essays by computer. *Phi Delta Kappan*, 47(5), 238–243.
- Parker, J. L., Becker, K., & Carroca, C. (2023). ChatGPT for automated writing evaluation in scholarly writing instruction. *Journal of Nursing Education*, 62(12), 721–727. <https://doi.org/10.3928/01484834-20231006-02>
- Prince, M. (2004). Does active learning work? A review of the research. *Journal of Engineering Education*, 93(3), 223–231. <https://doi.org/10.1002/j.2168-9830.2004.tb00809.x>
- Rahman, M. M., & Watanobe, Y. (2023). ChatGPT for education and research: Opportunities, threats, and strategies. *Applied Sciences*, 13(9), 5783. <https://doi.org/10.3390/app13095783>
- Rose, C. P., McLaughlin, E. A., Liu, R., & Koedinger, K. R. (2019). Explanatory learner models: Why machine learning (alone) is not the answer. *British Journal of Educational Technology*, 50(6), 2943–2958. <https://doi.org/10.1111/bjet.12858>
- Sarrion, E. (2023). Using ChatGPT to learn a language. In E. Sarrion (Ed.), *Exploring the power of ChatGPT: Applications, techniques, and implications* (pp. 117–128). Apress Media.
- Shah, R. A. (2020). Experiencing the process of knowledge creation: Use of inquiry-based learning in social work education. In I. Ponnuswami, & A. Francis (Eds.), *Social work education, research and practice* (pp. 43–57). Springer Nature.
- Shin, D., & Lee, J. H. (2023). AI-assisted automated scoring of picture-cued writing tasks for language assessment. *Education and Information Technologies*, 28, 7031–7063. <https://doi.org/10.1007/s10639-022-11473-y>
- Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research*, 78(1), 153–189. <https://doi.org/10.1002/j.2333-8504.2007.tb02053.x>
- Spronken-Smith, R. (2008). *Experiencing the process of knowledge creation: The nature and use of inquiry-based learning in higher education*.
- Spronken-Smith, R., & Walker, R. (2010). Can inquiry-based learning strengthen the links between teaching and disciplinary practice? *Studies in Higher Education*, 35(6), 723–740. <https://doi.org/10.1080/03075070903315502>
- Su, Y., Lin, Y., & Lai, C. (2023). Collaborating with ChatGPT in argumentative writing classrooms. *Assessing Writing*, 57. <https://doi.org/10.1016/j.asw.2023.100752>. Article 100752.
- Tegmark, M. (2017). *Life 3.0: Being human in the age of artificial intelligence*. Knopf.
- van den Bergh, H., & Rijlaarsdam, G. (2001). Changes in cognitive activities during the writing process and relationships with text quality. *Educational Psychology*, 21(4), 373–385. <https://doi.org/10.1080/01443410120090777>
- van den Bergh, H., Rijlaarsdam, G., & van Steendam, E. (2016). Writing process theory: A functional dynamic approach. In C. MacArthur, S. Graham, & J. Fitzgerald (Eds.), *Handbook of writing research* (2nd ed., pp. 57–71). The Guilford Press.
- Vandermeulen, N., Lindgren, E., Waldmann, C., & Levlin, M. (2024). Getting a grip on the writing process: (effective) approaches to write argumentative and narrative texts in L1 and L2. *Journal of Second Language Writing*, 65. <https://doi.org/10.1016/j.jslw.2024.101113>. Article 101113.
- Warschauer, M., & Ware, P. (2006). Automated writing evaluation: Defining the classroom research agenda. *Language Teaching Research*, 10(2), 157–180.
- Weaver, F. S. (Ed.). (1989). *Promoting inquiry in undergraduate learning. New directions for teaching and learning*. Jossey-Bass Inc.
- Yaghmaie, M., & Bahreininejad, A. (2011). A context-aware adaptive learning system using agents. *Expert Systems with Applications*, 38(4), 3280–3286. <https://doi.org/10.1016/j.eswa.2010.08.113>
- Yau, J. Y. K., & Joy, M. (2010). An adaptive context-aware mobile learning framework based on the usability perspective. *International Journal of Mobile Learning and Organisation*, 4(4), 378–390. <https://doi.org/10.1504/IJMLLO.2010.037535>
- Zare, J., Ai-Issa, A., & Madiseh, F. (2025). Interacting with ChatGPT in essay writing: A study of L2 learners' task motivation. *The Journal of the European Association for Computer Assisted Language Learning*, 37(3), 385–402. <https://doi.org/10.1017/S0958344025000035>
- Zhang, R., Zou, D., & Cheng, G. (2025). ChatGPT affordance for logic learning strategies and its usefulness for developing knowledge and quality of logic in English argumentative writing. *System*, 128, Article 103561. <https://doi.org/10.1016/j.system.2024.103561>
- Zhu, M., Zhang, M., & Gu, L. (2023). Insights into editing and revising in writing process using keystroke logs. *Language Assessment Quarterly*, 20(4–5), 445–468. <https://doi.org/10.1080/15434303.2023.2291478>
- Zou, M., & Huang, L. (2023). To use or not to use? Understanding doctoral students' acceptance of ChatGPT in writing through technology acceptance model. *Frontiers in Psychology*, 14, Article 1259531. <https://doi.org/10.3389/fpsyg.2023.1259531>
- Zou, M., & Huang, L. (2024). The impact of ChatGPT on L2 writing and expected responses: Voice from doctoral students. *Education and Information Technologies*, 29, 13201–13219. <https://doi.org/10.1007/s10639-023-12397-x>